

E073

Physics I Lab for Programmers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student tests whether pendulum length affects the time for one swing. She uses the same 50 g bob for every trial and codes the timer in Python. The lab room is kept at 21 °C. Which is the dependent variable?

- A. The time for one swing (period).
- B. The pendulum length.
- C. The room temperature.
- D. The 50 g bob mass.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A digital scale bought from an online store reads 1.52, 1.51, and 1.52 kg for a standard 1.00 kg mass. The lab's data logger saves each reading to a CSV file. How should the readings be described?

- A. Precise but not accurate, due to systematic error; recalibrate the scale.
- B. Random error; take more readings and average them.
- C. Accurate but not precise.
- D. An error in the CSV file format.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 1,500 kg test car starts from rest and accelerates at 3 m/s² for 4 s on a track built in 1998. How far does it travel?

- A. 48 m
- B. 12 m
- C. 7 m
- D. 24 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A cart speeds up from 10 m/s to 30 m/s in 5 s at a steady rate. What is its acceleration?

- A. 20 m/s²
- B. 6 m/s²
- C. 100 m/s²
- D. 4 m/s²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A 1,200 kg electric car, painted red and built in 2020, is pushed with a 3,000 N force while friction resists with 600 N. What is its acceleration?

- A. 3 m/s²
- B. 2 m/s²
- C. 2.5 m/s²
- D. 0.5 m/s²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A net force of 50 N gives an object an acceleration of 2 m/s². What is its mass?

- A. 25 kg
- B. 100 kg
- C. 52 kg
- D. 50 kg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A simulation, whose ball sprite is 32 pixels wide, starts with $x = 0$ m, $v = 2$ m/s, and $a = 4$ m/s². One step with $dt = 0.5$ s runs $v = v + a*dt$, then $x = x + v*dt$. What is x after the step?

- A. 4 m
- B. 1 m
- C. 2 m
- D. 1.5 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A 0.2 kg ball rolls off a 4.9 m high ledge at 3 m/s horizontally. The ball is bright orange. Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how far from the base of the ledge does it land?

- A. It cannot be found without knowing how the mass slows the ball.
- B. 3 m
- C. 14.7 m
- D. 4.9 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

From the same height, one ball is dropped straight down and another is launched horizontally at 5 m/s at the same moment. Ignoring air resistance, which lands first?

- A. The launched ball, because it is moving faster.
- B. The dropped ball, because the launched ball travels farther.
- C. They land at the same time.
- D. The heavier ball, always.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A 2 kg ball is dropped from 5 m in a lab with a new high-speed camera. Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how fast is it moving just before it hits the floor?

- A. 98 m/s
- B. 7 m/s
- C. 49 m/s
- D. About 9.9 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A velocity-time graph plotted with matplotlib shows a straight line from 2 m/s at $t = 0$ to 10 m/s at $t = 4 \text{ s}$. The student's laptop has 16 GB of RAM. What is the acceleration?

- A. 24 m/s^2
- B. 32 m/s^2
- C. 2 m/s^2
- D. 2.5 m/s^2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Using the same graph (2 m/s at $t = 0$ rising steadily to 10 m/s at $t = 4$ s), how far does the object travel?

- A.** 40 m
- B.** 24 m
- C.** 2 m
- D.** 8 m

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A student tests whether pendulum length affects the time for one swing. She uses the same 50 g bob for every trial and codes the timer in Python. The lab room is kept at 21 °C. Which is the dependent variable?

Target word: dependent variable **Decides:** The measured outcome is swing time.

A	Correct	It is what she measures in response.
B	Inverted operation	The variable she changes: independent.
C	Camouflage	A background condition, not the measured outcome.
D	Adjacent category	Held constant: a controlled variable.

2. Answer A Concept 2

A digital scale bought from an online store reads 1.52, 1.51, and 1.52 kg for a standard 1.00 kg mass. The lab's data logger saves each reading to a CSV file. How should the readings be described?

Target word: classify the error **Decides:** Readings agree closely but are all about 0.5 kg high.

A	Correct	Consistent offset = systematic error.
B	Confusion set	Averaging cannot remove a constant offset.
C	Inverted operation	The reverse: they are close together but far from true.
D	Camouflage	Storage does not change the readings.

3. Answer D Concept 3

A 1,500 kg test car starts from rest and accelerates at 3 m/s² for 4 s on a track built in 1998. How far does it travel?

Target word: distance under constant acceleration **Decides:** $x = \frac{1}{2}at^2$ with $v_0 = 0$.

A	Dropped element	Leaves out the $\frac{1}{2}$.
B	Adjacent category	That is the final speed in m/s, not the distance.
C	Inverted operation	Adds a and t instead of using the formula.
D	Correct	$\frac{1}{2} \times 3 \times 16 = 24$.

4. Answer D Concept 3 · transfer

A cart speeds up from 10 m/s to 30 m/s in 5 s at a steady rate. What is its acceleration?

Target word: solve for a **Decides:** $a = (v - v_0)/t$.

A	Adjacent category	The change in speed, not the rate.
B	Dropped element	30 / 5 ignores the starting speed.
C	Inverted operation	Multiplies by time instead of dividing.
D	Correct	20 / 5 = 4.

5. Answer B Concept 4

A 1,200 kg electric car, ~~painted red and built in 2020~~, is pushed with a 3,000 N force while friction resists with 600 N. What is its acceleration?

Target word: acceleration from net force **Decides:** $F_{\text{net}} = 3,000 - 600 = 2,400 \text{ N}$.

A	Inverted operation	Adds friction instead of subtracting it.
B	Correct	$2,400 / 1,200 = 2$.
C	Dropped element	Ignores friction: $3,000 / 1,200$.
D	Confusion set	Uses the friction force alone.

6. Answer A Concept 4 • transfer

A net force of 50 N gives an object an acceleration of 2 m/s^2 . What is its mass?

Target word: solve for m **Decides:** $m = F / a$.

A	Correct	$50 / 2 = 25$.
B	Inverted operation	Multiplies F by a.
C	Confusion set	Adds the numbers.
D	Wrong stem number	The force value.

7. Answer C Concept 5

A simulation, ~~whose ball sprite is 32 pixels wide~~, starts with $x = 0 \text{ m}$, $v = 2 \text{ m/s}$, and $a = 4 \text{ m/s}^2$. One step with $dt = 0.5 \text{ s}$ runs $v = v + a*dt$, then $x = x + v*dt$. What is x after the step?

Target word: trace one Euler step **Decides:** Velocity is updated first, then used for position.

A	Wrong stem number	The new velocity value, not position.
B	Adjacent category	Uses the old velocity, as if position were updated first.
C	Correct	$v = 2 + 4(0.5) = 4$; $x = 0 + 4(0.5) = 2$.
D	Confusion set	The exact kinematics formula, not what this code computes.

8. Answer B Concept 6

~~A 0.2 kg ball~~ rolls off a 4.9 m high ledge at 3 m/s horizontally. ~~The ball is bright orange.~~ Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how far from the base of the ledge does it land?

Target word: horizontal range **Decides:** Fall time $t = \sqrt{2h/g} = 1 \text{ s}$.

A	Camouflage	Mass does not affect projectile motion without air resistance.
B	Correct	$3 \text{ m/s} \times 1 \text{ s}$.
C	Confusion set	Multiplies speed by height.
D	Wrong stem number	The height of the ledge.

9. Answer C Concept 6 · transfer

From the same height, one ball is dropped straight down and another is launched horizontally at 5 m/s at the same moment. Ignoring air resistance, which lands first?

Target word: fall time comparison **Decides:** Fall time depends only on height.

A	Inverted operation	Horizontal speed does not speed up the fall.
B	Confusion set	Horizontal distance does not add fall time.
C	Correct	Both start with zero vertical velocity from the same height.
D	Absolute language	Mass does not affect fall time here.

10. Answer D Concept 7

~~A 2 kg ball is dropped from 5 m in a lab with a new high speed camera.~~ Ignoring air resistance and using $g = 9.8 \text{ m/s}^2$, how fast is it moving just before it hits the floor?

Target word: speed from energy conservation **Decides:** $v = \sqrt{2gh}$.

A	Dropped element	Forgets the square root.
B	Adjacent category	$\sqrt{gh}$ leaves out the factor of 2.
C	Confusion set	$g \times h$ is not a speed.
D	Correct	$\sqrt{2 \times 9.8 \times 5} = \sqrt{98} \approx 9.9$.

11. Answer C Concept 8

~~A velocity-time graph plotted with matplotlib shows a straight line from 2 m/s at $t = 0$ to 10 m/s at $t = 4$ s. The student's laptop has 16 GB of RAM.~~ What is the acceleration?

Target word: slope of v-t graph **Decides:** Slope = $(10 - 2)/4$.

A	Adjacent category	That is the area (displacement in m), not the slope.
B	Inverted operation	Multiplies Δv by time.
C	Correct	$8 / 4 = 2$.
D	Dropped element	$10 / 4$ ignores the starting speed.

12. Answer B Concept 8 · transfer

Using the same graph (2 m/s at $t = 0$ rising steadily to 10 m/s at $t = 4$ s), how far does the object travel?

Target word: area under v-t graph **Decides:** Area of a trapezoid.

A	Dropped element	10×4 assumes top speed the whole time.
B	Correct	$(2 + 10)/2 \times 4 = 24$.
C	Adjacent category	The slope (acceleration).
D	Confusion set	The change in velocity.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Independent, dependent, and controlled variables	The independent variable is what you change on purpose, the dependent variable is what you measure, and controlled variables are held constant.	Naming the changed variable as the dependent one.	“Does X affect Y?” in a lab setup.
2	Accuracy vs. precision; systematic vs. random error	Precision is how close repeated readings are to each other; accuracy is how close they are to the true value. A constant offset is systematic error and is not fixed by averaging.	Thinking more trials fix a calibration offset.	Tightly clustered readings that are all off by the same amount.
3	Constant-acceleration kinematics	$v = v_0 + at$ and $x = v_0t + \frac{1}{2}at^2$.	Dropping the $\frac{1}{2}$ or reporting velocity when distance is asked.	Starting speed, acceleration, and time given.
4	Newton’s second law uses net force	$a = F_{\text{net}} / m$, where F_{net} is the vector sum of all forces.	Using only the applied force and ignoring friction.	An applied force plus an opposing force.
5	Euler time-stepping in simulations	Each step updates velocity from acceleration and position from velocity over a small dt ; the order of updates and the size of dt change the result.	Expecting the code to match the exact formula, or ignoring update order.	A loop with $v = v + a*dt$ and $x = x + v*dt$.
6	Projectile motion: independent axes	Horizontal velocity stays constant (no air resistance) while vertical motion falls freely; fall time depends only on height.	Thinking horizontal speed changes fall time.	Something launched sideways off a ledge.
7	Conservation of energy	Without friction, $mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh}$ and mass cancels.	Forgetting the square root or thinking heavier objects land faster.	An object dropped from a height and a speed asked.
8	Reading motion graphs	On a velocity–time graph, slope is acceleration and area under the line is displacement.	Mixing up slope and area.	A v – t graph and “how far” or “how fast is it speeding up.”

Course scope (public description): This course introduces foundational physics through programming-based lab experiences. Students apply the scientific method, analyze experimental data, and communicate results while modeling physical systems. Through applied projects, learners create physics-based simulations and computational artifacts, building practical skills in scientific reasoning, data analysis, and technical communication.